# RAMCO INSTITUTE OF TECHNOLOGY

**Department of Computer Science and Engineering**
**Academic Year: 2019-2020(Even Semester)**

**Degree, Semester& Branch: II Semester B.E. CSE**
**Course Code & Title: CS8251 Programming in C**
**Name of the Faculty member: Dr.K.Vijayalakshmi & Mrs.B.Vijayalakshmi**
**Topic: Decision Making Statement and Looping statements**
**Practice's followed: Code Debugging**
**Date: 21.01.2020**

**Objectives of the activity:**

• To enable the students to familiarize the concepts of decision making and looping statements and improve the programming skills.

**Justification for choosing the topic:**

Decision making and looping statement are the basic building blocks for writing a program. Code Debugging makes students to analyse, think and come out with more number of solutions to run the given code without bugs. In order to improve the analytical skills of a student debugging activity is carried out in the topic.

**Activity:**

Students were asked to form a team of two members for this activity. In this activity, each team was given with 2 C programs which contains a set of partially completed and error prone code along with a desired output. The incomplete code may for finding palindrome number, sum of digits, finding factors of a number and identifying vowel or consonant.

The activity was conducted during the last 20 minutes of the class. The students have to think and identify the bugs present in the code, rectify it and complete the missed part in the program. Also the students have to verify the logic in order to get the required output. After completing the code, every student from a team discussed their code and the findings to all other team members.

**Observations made:**

The completed code was corrected during the hour itself and the following observations were made.

- Some of the students find very easy to complete the code and performed debugging easily.
- Some of them need more clarification on the syntax of looping statement.
- Additional examples shall be given them to make understand the concept easily.

**Outcome:**

This activity was very useful to identify whether the students understand the concept of decision making and looping statement. Through this activity, the students are able to identify the errors and perform error correction easily in order to execute program correctly.

**Relevance Mapping**

This activity helps in attaining the following Objectives – PO mapping:

| Objectives / PO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **O1** | **3** | **3** | **3** | **2** | **-** | **-** | **-** | **-** | **3** | **3** | **-** | **1** |

**Pre-implementation:**

**Identified Challenges:**

- Some of the students feel easy to complete the missed part of a code.
- Some of them feel difficult to identify the syntax error since they were not familiar with the syntax.
- Some of the students who were not interested in programming were not actively participated.

**Addressing challenges**

- I explained the concepts again with the syntax of different programming constructs.
- The students were motivated to have the better coding practice

**Post implementation**

- Students can understand the working of different looping and decision making statement.
- It was a different experience for them to learn the inner working of a code in order to get the desired output, which creates interest to them for writing programs.

References:


1. https://www.ritrjpm.ac.in/images/computer-science/26_OIT551_Colloborativecoding.pdf
2. https://www.ritrjpm.ac.in/images/computer-science/18_EC8393_Codedebug.pdf

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**Team Members:**

**Identify the error and complete the code to get the desired output.**

1.
```
#include <stdio.h>
int main()
{
char c;
printf("Enter a character");
scanf "%c", &c );
if(c == 'a' || c == 'e' || c == 'i' || c == 'o' || c == 'u' || c == 'A' || c == 'E' || c == 'I' ||
  c == 'O' || c == 'U')
{ printf ~~scanf~~("%c is a vowel.", c);
}
else
    printf("%c is a consonant.", c );
return 0;
}
```

**Output:**
**Enter a character e**
**e is a vowel**

2.
```
#include <stdio.h>
int main()
{ int ~~char~~ n, i;
printf("Enter an integer: ");
scanf("%d", &n);
for (i=1; i <=5; ++i)
{
  printf("%d * %d = %d \n", n, i, (n*i) );
}
return 0;
}
```

**Output:**
**Enter an integer: 6**
**6 * 1 =6**
**6 * 2 = 12**
**6 * 3 = 18**
**6 * 4 = 24**
**6 * 5 = 30**

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**Team Members:**

**Identify the error and complete the code to get the desired output.**

```
1.include <stdio.h>
 int main()
 {
  int num;
  printf("Enter a number: ");
  scanf("%d", &num )
  if (num <= 0 )
  {
    if (num == 0.0);
      printf( "You entered zero" );
    else
      printf("You entered a negative number.");
  }
  else
    printf("You entered a positive number.");
  return 0;
 }
```

**Output:**
**Enter a number: -5**
**You entered a negative number.**

```
2.#include <stdio.h>
 int main()
 {
  int n;
  int count=0;
  printf("Enter an integer: ");
  scanf("%d",&n);
  while ( n>0 )
  {
    n /= 10;
    count++ ;
  }
  printf("Number of digits: %d", count);
 }
```

**Output:**
**Enter an integer: 2589**
**Number of digits: 4**

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**Approved By**

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